

Spiros Kontopoulos

Senior Software Engineer | Python | Backend & AI Systems

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🐙 github.com/spiros-kontopoulos-dev | Remote / Hybrid / On-site · Open to relocation

PROFILE

Backend-focused senior software engineer with **10+ years** of experience building **data-intensive systems at scale**, combining **Python, SQL, backend architecture, and distributed data workflows**. Technical owner of **location-data systems** at Adzuna, a global job platform processing **tens of millions of ads across 20+ markets**.

Currently building Codiquiz, an AI-powered coding quiz platform with a **RAG-based knowledge service** for grounded content generation, retrieval, quality control, and technical verification.

Strong focus on **system design, domain modelling, AI reliability, and maintainable software** that solves real business problems, with proven experience delivering independently in remote environments.

EDUCATION & CERTIFICATIONS

BSc in Informatics Engineering, Technological Institute of Crete 2002 – 2008

Thesis: “Web-based food ordering application” - backend architecture and development (PHP, MySQL, AJAX)

Internship: Hellenic Telecommunications Organization (OTE) - IT & Network Operations Intern

Professional Development: Python 3 Programming Specialization – University of Michigan, Coursera (118h)

SKILLS

Backend Core: Python (5+ yrs), SQL (10+ yrs), FastAPI (3+ yrs), Pytest

AI Engineering (1+ yrs): OpenAI API, Prompt Engineering, RAG, LangChain, LangGraph, Hugging Face, Vector Search

Architecture: System Design, ORM, API Design, REST APIs, GraphQL, Distributed Systems

Async, Messaging & Orchestration (~2 yrs): Redis, Celery, Apache Airflow, RabbitMQ

Databases & Search: PostgreSQL, MySQL, pgvector, Solr, Elasticsearch, Amazon Athena

Cloud & Infrastructure: Docker, Git, CI/CD, Jenkins, Kibana, Linux, AWS S3, EC2

Frontend: React/TypeScript/Tailwind CSS (2+ yr), JavaScript/AJAX/HTML/CSS (5+ yrs), Playwright

Additional Languages & Frameworks: Perl, PHP, Django, Laravel

Languages Spoken: Greek (native), English (professional working proficiency)

PROFESSIONAL EXPERIENCE

Founder & Senior Python/AI Engineer - Codiquiz (AI-Powered Coding Quiz Platform) Jan 2026 – Present

Python, FastAPI, PostgreSQL, pgvector, RAG, OpenAI API, LangChain, LangGraph, Hugging Face, Celery, Redis, Airflow, Docker, Pytest, React, TypeScript, TailWind CSS, Playwright

preview.codiquiz.com · Admin Demo Access · GitHub & Docs

Building an end-to-end programming question-bank platform, designed beyond a simple AI wrapper with layered backend/AI systems for planning, generation, validation, quality control, and admin review.

- **Platform Architecture:** Designed a Dockerised microservices-style architecture with dedicated API, generation, worker, scheduler, database, and caching services.
- **Taxonomy Model:** Designed a structured knowledge model for programming technologies (currently Python), organising content into domains, topics, and **3,000+ granular testable concepts** for systematic coverage tracking.
- **Content Coverage Engine:** Built a coverage engine that detects gaps across topics, difficulty levels, and question formats, using scoring and mapping rules to prioritise what the AI should generate next.
- **AI Generation Pipeline:** Orchestrated workflows with prompt engineering, response validation, staging, admin review/approval, OpenAI Batch API, retries, and token/cost tracking, enabling controlled AI generation.
- **Quality Control Systems:** Built taxonomy-based classification, duplicate detection, semantic similarity warnings, pattern metadata, and review tooling for generated content, improving review safety and reducing repeated questions.
- **Workflow Automation:** Designed and integrated Celery background jobs and Airflow-orchestrated workflows for scalable question generation, batch processing, retries, validation, and review.

- **RAG Knowledge Service:** Built a standalone FastAPI service using Hugging Face Sentence Transformers, PostgreSQL/pgvector, and OpenAI to ingest official programming documentation, retrieve concept evidence, ground question generation, and verify generated content against authoritative sources.

Software Engineer, Adzuna – Remote (London, UK)

Mar 2019 – Mar 2026

Global job-search platform processing tens of millions of job ads across 20+ international markets.

Backend Engineer → Senior Software Engineer (Python, Perl, Javascript)

2022 – 2026

- **Technical Ownership & Leadership:** Owner of the location data domain, leading technical decisions and designing backend systems and complex data workflows for large-scale location ingestion, search, and data quality management.
- **Technical Mentoring & Code Reviews:** Provided code reviews, onboarding support, and technical guidance for developers working on feed integrations, XML/HTML ingestion, and data-processing workflows.

Backend Key Achievements

- **Search Optimisation (Solr):** Resolved Apache Solr search inconsistencies affecting **1M+ job listings** by analysing hierarchical ranking edge cases. Delivered safe backend fixes for location normalisation (~**850K ads**) and Solr configuration updates for company alias issues (~**150K ads**), documenting fixes to reduce future debugging.
- **AI-Assisted Location Data Validation:** Improved location data quality by **8%** by building a hybrid validation framework combining external geocoding APIs, a custom confidence-scoring algorithm, and LLM-based analysis of ambiguous cases to identify invalid and misaligned location records at scale.

Selected End-to-End / Full-Stack Systems

- **Locations Management System:** Designed a real-time editing system replacing traditional CRUD workflows with staged edits and preserving edits across navigation and pagination. Introduced new data models and backend logic using **structured action payloads** to apply batched location modifications.
- **Admin Data Access Optimisation:** Redesigned the admin job-ads browsing system for complex EAV-schema data, replacing inefficient per-field queries with a single optimised view layer, improving performance by at least **3–4x** and improving scalability.
- **Feed Configuration System:** Redesigned from error-prone CSV updates to a unified admin workflow with event tracked change history, enabling PM self-service and reducing manual configuration effort by ~**50%**.

Backend Developer → Backend Engineer (Perl)

2019 – 2022

- **Location Data Pipeline:** Built an automated geocoding pipeline to ingest, validate, normalise, and enrich US location data, onboarding **1,000+ locations** and expanding coverage by ~**2.5%**.
- **Scraping Framework Redesign:** Collaborated on redesigning the internal Perl scraping and source-management framework for XML/HTML feed integrations, improving maintainability by **30%**.
- **Click Tracking & Analytics:** Extended click-tracking and reporting workflows over large datasets using dynamic backend logic and Amazon Athena queries, enabling flexible client analytics and custom reporting scenarios.

Full Stack Developer (PHP), IZIBIT - Zakynthos, Greece

Feb 2013 – Jan 2015

Technology company providing an online advertising platform for tourist businesses, including hotels and restaurants.

- Developed full-stack functionality across PHP, MySQL, and AJAX-driven frontend systems, delivering new features and enhancements across the platform.
- Designed and delivered internal administration and business-onboarding tools that streamlined operations and reduced manual processing effort by approximately **50%**.

Second Lieutenant – IT Office Director, Hellenic Army - Tripoli, Greece

2011 – 2012

- Served as IT Office Director during compulsory military service, overseeing technical support, system operations, and administrative coordination; discharged at the rank of Second Lieutenant.

SELECTED INDEPENDENT DEVELOPMENT

- **Web Data Extraction Tool (Perl, 2018):** Perl scraping pipeline for extracting, parsing, normalising, and exporting web data, HTTP requests, HTML parsing, regex, retry handling, and structured output generation.
- **Music Library Organizer (Python, React, 2017):** Converted into desktop application using Electron to organise and classify local music files, extracting metadata and integrating external APIs to enrich missing information.
- **Online Food Ordering Platform (Python, Django, 2016):** Rebuilt and extended a web-based food ordering system, implementing backend logic, database design, and API endpoints to manage orders, users, menu data.